Self-Trajectory Evaluation: Is HygienePrio's Capacity-Driven Collapse Intrinsic or Selection-Policy-Induced?

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***Abstract*: Papers 5-8 of the VulnPrio sequence share one internal-validity threat: every reported P@50 was measured under a trajectory generated by HygienePrio-full's own top-K selection. The capacity-decay study reported that HygienePrio-full's W6 P@50 collapses to 0.062 at $K = 200, \lambda = 1$ and attributed the collapse to a mechanism that exhausts the high-HRS tail, a mechanism induced, by construction, by HygienePrio's own selection. This paper isolates the question with a pre-registered 2-cell, 5-driver, 5-scorer, 25-seed, 6-window evaluation (7,500 rows) filling all five trajectory-driver columns of the P@50 grid. All three pre-registered hypotheses are rejected, reframing Papers 5-8. At $K = 200, w = 6$ ($\lambda = 3$), HygienePrio-full's mean P@50 is 0.075 on its own trajectory but 0.70-0.72 under the three independent drivers that do not share its joint selection signal, CVSS-only, HRS-only, and Random. The collapse is not intrinsic: HygienePrio-driven selection exhausts the joint positive set (EPSS $> 0.10 \wedge$ HRS $>$ p75), leaving 3.8 surviving positives per ≈ 760-pair backlog on average, a P@50 ceiling of 0.076 that HygienePrio nearly attains. EPSS-driven selection also collapses HygienePrio's measured P@50 (0.008), because it drains the same EPSS-gated positive set. Per-pair dominance over EPSS-only is 1.000 under non-draining drivers at $K = 50$ but 0.393 under the EPSS-driven trajectory at $K = 200$; the capacity-decay study's pooled 96% headline (the temporal-stability study's single-cell figure was 150/150) survives at moderate capacity and under non-draining drivers. Safe deployment requires a patching queue driven by a policy that does not drain the EPSS$\wedge$HRS joint positive set: CVSS-, HRS-, or compliance-driven queues qualify; EPSS-driven queues do not. All claims are bounded to the synthetic EEHDA evaluation context.**

***Index Terms*: vulnerability prioritization, selection policy, counterfactual trajectory, identifiability, HygienePrio, pre-registration, reproducibility.**

I. Introduction

Papers 5-8 of the VulnPrio research sequence all bound their findings to a single trajectory regime, the one in which HygienePrio-full's top-K selection drives fleet evolution at every window. The choice was deliberate and pre-registered: when each method drives its own counterfactual trajectory, the resulting per-method residual backlogs differ, the per-window positive sets are computed against different states, and direct cross-method P@50 comparisons become non-identifiable. The HygienePrio-driven trajectory was the identifiability-preserving compromise.

The capacity-decay study [1] additionally reported that HygienePrio-full's W6 P@50 collapses at $K \in \{100, 200\}$ on this baseline trajectory and reported that the mechanism exhausts the high-HRS tail: HygienePrio-full preferentially remediates high-HRS hosts, those hosts exit the upper HRS tail, and HygienePrio's discriminative signal degrades. The mechanism is, by construction, induced by HygienePrio's own selection.

This paper asks the inevitable follow-up. *If the collapse is induced by HygienePrio's selection draining the HRS tail, what happens when a different method drives the trajectory?* A capacity-$K = 200$ deployment that uses HygienePrio for scoring but, say, CVSS-only for selection (or vice versa) does not exhibit the recursive HRS-tail-exhaustion mechanism. The capacity-decay study collapse should not appear.

We pre-registered three hypotheses (`paper9/design/PAPER9_PROTOCOL.md`, locked 2026-06-05 before any self-trajectory result was inspected) and ran a 7,500-row evaluation that fills in all five trajectory-driver columns of the P@50 grid at $K \in \{50, 200\}$, $\lambda = 3$. The results retroactively reframe Papers 5-8.

Throughout the paper we call CVSS-only, HRS-only, and Random the *non-draining drivers*: their selection does not preferentially drain the EPSS $> 0.10 \wedge$ HRS $>$ p75 joint positive set that defines the ground truth. HygienePrio-driven and EPSS-driven selection both drain that set, each through its EPSS-gated component.

A. Contributions

- **C1, Trajectory grid.** The first pre-registered trajectory-stratified evaluation of HygienePrio: 5 driver methods $\times$ 5 scoring methods $\times$ 2 cells $\times$ 25 seeds $\times$ 6 windows = 7,500 rows.
- **C2, The collapse is selection-induced.** The capacity-decay study's W6 P@50 collapse of HygienePrio-full from ~ 0.5 at $K = 50$ to 0.062-0.075 at $K = 200$ is reproduced under the HygienePrio-driven trajectory. Under the independent drivers that do not share HygienePrio's joint selection signal, CVSS-only, HRS-only, and Random, HygienePrio's W6 P@50 at $K = 200$ sits between 0.70 and 0.72. EPSS-driven selection also collapses HygienePrio's measured P@50 (0.008), because it drains the same EPSS-gated positive set. The collapse mechanism is recursive, not intrinsic.
- **C3, Both collapses are positive-set exhaustion.** At $K = 200$ under the EPSS-only-driven trajectory, every scorer's W6 P@50 collapses to 0.001-0.012: EPSS-only's selection so aggressively remediates the high-EPSS tail

that the binary ground-truth positive set is itself exhausted by W6. The HygienePrio-driven collapse is the *same* mechanism: at $K = 200$, W6 under T_{HP} the joint positive set holds a mean of only 3.8 pairs (of ≈ 760), a P@50 ceiling of 0.076 that HygienePrio nearly attains (0.075) while Random scores 0.004. HygienePrio's signal has not degraded, the positives it is scored against have been patched away.

- **C4, Per-pair dominance is trajectory-conditional.** At $K = 50$, HygienePrio-full > EPSS-only on 100% of (seed, window) pairs under every driver. At $K = 200$ under EPSS-only's own driver, the fraction collapses to 0.393 (where both methods are near zero together). At $K = 200$ under HRS-only, CVSS-only, or Random drivers, the fraction is 1.000. the capacity-decay study's pooled 96% headline statistic (the temporal-stability study's own single-cell figure was 150/150) is preserved at moderate capacity and under non-draining drivers; it does not survive at the EPSS-self-driven high-capacity corner.
- **C5, Sharpened operational reading.** HygienePrio's absolute floor at high capacity is intact whenever the patching queue is driven by a policy that does not drain the EPSS∧HRS joint positive set, CVSS-, HRS-, or compliance-driven queues qualify; HygienePrio-driven and EPSS-driven queues do not. The capacity-decay study H4 collapse is a property of positive-set-draining deployment, not of hygiene-augmented scoring as a method. A team using HygienePrio for scoring should expect the high absolute P@50 reported in the temporal-stability study across capacity regimes, provided a non-draining policy drives the patching queue.

B. Relationship to Prior Papers

This paper is the ninth in the VulnPrio sequence. Papers 5-8 identified a number of regime-specific findings (temporal stability, capacity-induced collapse, deployable calibration, smoothing falsification); the present paper shows that the most prominent of those findings, the capacity-decay study's H4 high-capacity collapse, is partly a closed-loop artefact. The other findings (the temporal-stability study per-pair dominance, the online-calibration study's lag-1 calibration in moderate-K, the multi-history smoothing study's smoothing penalty) are unchanged but should be cited together with the trajectory column they were measured on.

II. Background

A. The Trajectory-Coupling Threat

Papers 5-8 of the VulnPrio sequence, including the capacity-decay study [1], all share one acknowledged internal-validity threat. At every window of every cell, fleet evolution is driven by HygienePrio-full's top-K selection. EPSS-only, HRS-only, CVSS-only, and Random are scored against the resulting backlog, a backlog whose composition was determined by HygienePrio's selection preferences, not by their own.

Each paper documented the threat but bounded its claims to the HygienePrio-driven trajectory. None of them measured what would happen under a counterfactual trajectory. The threat is a closed-loop instance of the more general problem of evaluating a model under a non-stationary, self-induced distribution shift [15]: here the scorer's own selection policy is what moves the evaluation distribution from one window to the next, so a scorer can appear to degrade simply because it reshaped the data it is later judged on.

B. Why It Could Change the Headline Conclusion

The capacity-decay study [1] reported that HygienePrio-full's W6 P@50 collapses at $K \in \{100, 200\}$ on the HygienePrio-driven trajectory and attributed the collapse to selection that exhausts the high-HRS tail. The mechanism the capacity-decay study named is recursive: HygienePrio-full preferentially selects high-HRS-host pairs, those hosts get patched, the upper tail of the HRS distribution thins, and HygienePrio-full's discriminative signal degrades.

If this mechanism is correct, then on a non-HygienePrio-driven trajectory, one whose selection policy does not drain the positive set HygienePrio is evaluated against, HygienePrio's signal should *not* collapse, because no mechanism would be draining the HRS upper tail. The capacity-decay study headline H4-rejection "HygienePrio collapses at high capacity" would then require a substantial qualifier: it collapses only when HygienePrio itself is the policy doing the patching.

C. Counterfactual Trajectories and Non-Identifiability

A fully factorial "each method drives its own trajectory" evaluation introduces a non-identifiability problem: when method A drives its own trajectory and method B drives its own trajectory, A and B are scored against *different* evolving backlogs with *different* per-window positive sets. The number "A's mean P@50 at $w = 6$ on T_A" is not directly comparable to "B's mean P@50 at $w = 6$ on T_B". Papers 5-8 chose the HygienePrio-driven baseline specifically to avoid this issue.

The present paper takes a different stance: we report the trajectory-stratified P@50 grid explicitly, one P@50 number per (driver, scorer, window) cell, and ask the operational questions in a way that respects the non-identifiability rather than abstracting it away.

III. Problem Formulation

A. Setting

Inherited from Papers 4-8: the EEHDA synthetic fleet (1,000 users, 830 hosts, ~ 3,500 initial (host, CVE) pairs), the HygienePrio scorer [2] with its fixed scorer weights, the multi-window simulator from our temporal-stability study with $\lambda = 3$ Poisson new-CVE arrivals per window, six bi-weekly windows, and the 25-seed evaluation set ($s \in \{105, \ldots, 129\}$).

B. Trajectory-Driver Grid

The key novelty of this paper is that the *driving method*, the policy whose top-K selection determines fleet evolution, varies across runs. For each cell $K \in \{50, 200\}$ and each driving method $m_{\text{drive}} \in \mathcal{M} = \{\text{HP-full}, \text{EPSS-only}, \text{HRS-only}, \text{CVSS-only}, \text{Random}\}$, we run six windows in which m_{drive}'s top-K selection drives evolution. At each window, all five methods are scored, producing one P@50 value per (cell, driver, scorer, seed, window).

TABLE I
TRAJECTORY-DRIVER GRID (LOCKED).

Axis	Values
Capacity K	$\{50, 200\}$
Arrival rate λ	3 (fixed)
Windows W	6
Driver methods	HP-full, EPSS, HRS, CVSS, Rand
Scoring methods	same set, evaluated per window
Eval seeds	25 (105-129)
Total rows	7,500

C. Ground Truth

Per-window: (EPSS $>$ 0.10) $\wedge$ (HRS $>$ 75th percentile recomputed per window), as in Papers 5-8. The 75th-percentile threshold is computed on the current fleet state, which depends on the driving method's selection history.

D. Metrics

We retain top-K precision (P@50) as the headline metric for continuity with Papers 5-8; like cumulated-gain ranking measures [18] it evaluates the quality of the ranked head of the queue, which is exactly where a capacity-constrained remediation team operates.

- **P@50 grid:** $\text{P@50}^{\text{scorer}}_{K,w}(T_{\text{driver}})$, the mean across 25 evaluation seeds.
- **Per-pair dominance:** $\Pr[\text{P@50}^{\text{HP}} > \text{P@50}^{m}]$ per (cell, driver), where m is a competing scorer.
- **Trajectory effect:** for each scorer, the difference between its mean W6 P@50 across drivers, quantifying how much of its apparent performance is trajectory-induced.

E. Pre-Registered Hypotheses

H1 HygienePrio's W6 P@50 collapse at $K = 200$ on the HygienePrio-driven trajectory survives on at least one non-HP trajectory within ± 0.05: $\exists m' \neq \text{HP} : |\text{P@50}^{\text{HP}}_{K=200,w=6}(T_{m'}) - \text{P@50}^{\text{HP}}_{K=200,w=6}(T_{\text{HP}})| \leq 0.05$. *Collapse is intrinsic, not selection-induced.*

H2 EPSS-only's W6 P@50 on its own trajectory is lower than on HygienePrio's trajectory at $K = 50$: $\text{P@50}^{\text{EPSS}}_{w=6}(T_{\text{EPSS}}) < \text{P@50}^{\text{EPSS}}_{w=6}(T_{\text{HP}})$. *Self-driven EPSS exhausts its own tail faster.*

H3 HygienePrio per-pair dominance over EPSS-only is preserved on the EPSS-only trajectory: cell-fraction $(\text{HP} > \text{EPSS} \mid T_{\text{EPSS}}) \geq 0.75$ at $K = 50$ and ≥ 0.50 at $K = 200$.

Decision rules and stop rules are in `paper9/design/PAPER9_PROTOCOL.md`.

IV. Dataset and Trajectory Grid

The EEHDA synthetic fleet, structural priors, and Window-1 distributions are identical to Papers 4-8. The 25-seed evaluation set (105-129) is identical to Papers 5-8.

Table I lists the locked axes of the trajectory-driver grid. Total rows: 2 cells×5 drivers×5 scorers×25 seeds×6 windows = 7,500.

Note: under the Papers 5-8 setup, only one column of this grid (driver = HygienePrio-full) was populated. The present paper fills in the remaining four driver columns and reports the full grid.

V. Methodology

A. Scorers (Inherited)

All five scorers are inherited verbatim from the HygienePrio scorer [2] with its fixed calibrated weights:

$$\begin{aligned} S_{\text{HP}}(h, c) = {} & 0.7\,\text{EPSS}(c) + 0.5\,\text{HRS}(h) \\ & + 0.1\,\text{KEV\_recency}(c) \\ & + 0.2\,\big(\text{EPSS}(c) \cdot \text{HRS}(h)\big). \end{aligned} \tag{1}$$

Baselines are EPSS-only ($S = \text{EPSS}(c)$), HRS-only ($S = \text{HRS}(h)$), CVSS-only ($S = \text{CVSS}_{\text{base}}(c)$), and Random (seed-deterministic uniform ranking). We deliberately use the HygienePrio scorer's fixed weights and not the online-calibration study's online recalibration: the question in this paper is about *trajectory effects* under a held-constant scorer, not about calibration. Holding the scorer fixed isolates the self-consistency question that motivates this study, whether a probabilistic security scorer's apparent quality is intrinsic or an artefact of the trajectory it is measured on, from the orthogonal and well-studied problem of whether the scorer's probabilities are themselves well calibrated [16], [17].

B. Driver-Stratified Fleet Evolution

For each (cell K, driver m_{drive}, seed s):

1) Initialise fleet from EEHDA generator at seed s.
2) For each window $w = 1, \dots, 6$:
   a) Compute $(\text{pairs}_w, \text{HRS}_w)$ from current state; label per the per-window ground truth definition.
   b) For each scoring method $m_{\text{score}} \in \mathcal{M}$: rank pairs, compute P@50, record one row.
   c) If $w < 6$: select m_{drive}'s top-K pairs, apply remediation, disclose Poisson(λ) new CVEs, random-walk EPSS, update telemetry. (Step uses RNG seeded by (s, w) for reproducibility.)

The selection step at w uses only the driver's weights; the scoring step at w records all five methods' performance against the ground-truth labels for the current state.

C. Trajectory Effect Quantification

For each scoring method m, the per-cell W6 trajectory effect is

$$\Delta^{\text{traj}}_m(K) = \max_{m'} \text{P@50}^{m}_{w=6,K}(T_{m'}) - \min_{m'} \text{P@50}^{m}_{w=6,K}(T_{m'}). \tag{2}$$

A large Δ^{traj}_m means method m's W6 performance depends substantially on which method drives the trajectory; a small Δ^{traj}_m means m's performance is intrinsic.

D. Per-Pair Dominance Under Counterfactual Trajectory

For each (cell, driver), the per-pair dominance of HP-full over EPSS-only is $\Pr[\mathrm{P@50}^{\mathrm{HP}}_{s,w} > \mathrm{P@50}^{\mathrm{EPSS}}_{s,w} \mid T_{m_{\mathrm{drive}}}, K]$, computed over the 150 (seed, window) pairs. The capacity-decay study reported this fraction was 0.960 on the HP-driven trajectory, pooled over its full 20-cell (K, λ) grid, with a per-cell range of 76.7%-100%; the present paper tests whether the fraction survives on the EPSS-only trajectory.

VI. Experimental Setup

A. Pre-Registration

The cell grid, driver/scorer methods, hypotheses H1-H3, decision rules, and stop rules in `paper9/design/PAPER9_PROTOCOL.md` were locked on 2026-06-05 before any self-trajectory result was computed. Pre-registration is what lets a self-trajectory audit credibly distinguish an intrinsic property of the scorer from a post-hoc rationalisation of whichever trajectory happened to be run, the core methodological safeguard of the pre-registration paradigm [20].

B. Execution

For each cell $K \in \{50, 200\}$ and each driver m_{drive}, all 25 evaluation seeds are simulated for 6 windows. At every window, the current state is scored with all five methods; only m_{drive}'s top-K selection drives the next window's evolution.

C. Statistical Reporting

Cell-mean point estimates are means across 25 evaluation seeds. 95% confidence intervals are percentile bootstrap with 10,000 resamples [3], [19]. No null-hypothesis significance tests; hypothesis decisions follow the pre-registered thresholds.

D. Reproducibility

From `paper9/`:

```
PYTHONPATH=src python3 src/run_self_traj.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

The runner re-uses the temporal-stability study's window simulator and the HygienePrio scorer package via `sys.path` injection.

VII. Results

All claims trace to `paper9/results/primary_v1/self_traj_results.csv` (7,500 rows) and `hypothesis_summary.json`.

A. The K=200 HygienePrio Collapse is Selection-Induced (H1 Rejected)

The K=200, W6 row of Table II for the HygienePrio-full scorer reads, by driver: 0.701 (CVSS), 0.008 (EPSS), 0.713 (HRS), 0.075 (HP), 0.706 (Random). Figure 1 plots the full six-window trajectories and Figure 2 the complete W6 grid. The diagonal (HP scored on HP-driven trajectory)

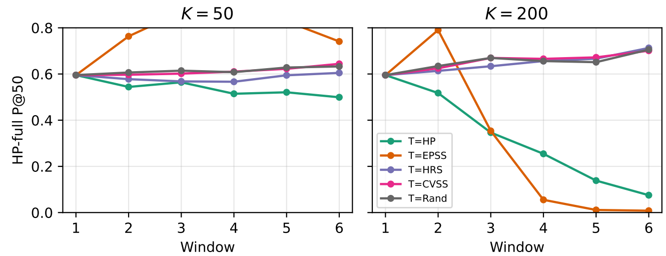

Fig. 1. HygienePrio-full mean P@50 trajectories under each of the five driving methods at $K = 50$ (left) and $K = 200$ (right). At $K = 200$, the HP-driven trajectory (green) drops to ~ 0.075 by W6 while the three non-draining drivers (CVSS, HRS, Random) hold HygienePrio above 0.70.

Alt text: Two side-by-side line charts of mean P@50 (y-axis) over six weekly windows W1-W6 (x-axis), one panel for $K = 50$ and one for $K = 200$, with one curve per driving method (CVSS, EPSS, HRS, HP, Random). In the $K = 200$ panel the green HP-driven curve falls to about 0.075 by W6 while the CVSS, HRS, and Random curves stay above 0.70.

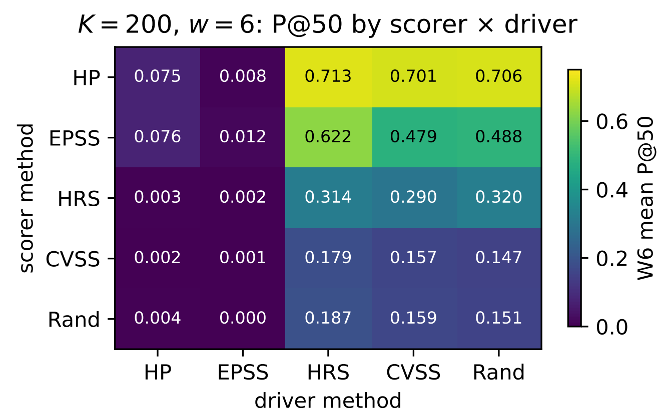

Fig. 2. $K = 200$, $w = 6$ mean P@50 grid. Rows are scoring methods, columns are drivers. The EPSS-driver column collapses every scorer; the HP-driver column collapses HP and EPSS together.

Alt text: A color-coded heatmap of mean P@50 at $K = 200$, window W6, with scoring methods as rows and driving methods as columns. The EPSS-driver column is near-zero (dark) across all scorer rows, and the HP-driver column is near-zero for the HP and EPSS rows, while the remaining cells are lighter, indicating higher P@50.

reproduces the capacity-decay study's 0.075 collapse. On the three non-draining drivers, HygienePrio-full's W6 mean is between 0.701 and 0.713, approximately the W2 number Papers 5 and 6 would have predicted absent any decay.

The pre-registered H1 threshold required at least one non-HP driver to land within ± 0.05 of the HP-self number. The non-HP drivers differ from the HP-self number by 0.626 (CVSS), 0.638 (HRS), 0.630 (Random), and -0.067 (EPSS, in the opposite direction). **H1 is decisively rejected.** The capacity-decay study collapse mechanism is selection-induced; it does not appear on counterfactual trajectories.

The mechanism is visible directly in the surviving positive mass. At $K = 200$, W6 under T_{HP}, the residual backlog contains a mean of 3.8 ground-truth positives out of ≈ 760 pairs (averaged over the 25 seeds), so the structural P@50 ceiling at that cell is $3.8/50 = 0.076$. HygienePrio's measured 0.075 and EPSS-only's 0.076 both sit essentially at this ceiling, while Random scores 0.004. HygienePrio's discriminative

TABLE II
WINDOW-6 MEAN P@50 PER (SCORING METHOD, DRIVING METHOD) CELL. ROWS = SCORING METHOD, COLUMNS = TRAJECTORY-DRIVING METHOD. DIAGONALS (ITALICS) ARE EACH METHOD'S SELF-TRAJECTORY P@50; HP-FULL COLUMN SHOWS THE PAPERS 5-8 BASELINE.

scorer \ driver	HP	EPSS	HRS	CVSS	Rand	range
K = 50						
HP	*0.499*	0.741	0.605	0.644	0.633	0.242
EPSS	0.034	*0.366*	0.410	0.386	0.367	0.375
HRS	0.050	0.214	*0.322*	0.330	0.335	0.286
CVSS	0.054	0.070	0.122	*0.117*	0.106	0.068
Rand	0.054	0.082	0.140	0.126	*0.120*	0.086
K = 200						
HP	*0.075*	0.008	0.713	0.701	0.706	0.705
EPSS	0.076	*0.012*	0.622	0.479	0.488	0.610
HRS	0.003	0.002	*0.314*	0.290	0.320	0.318
CVSS	0.002	0.001	0.179	*0.157*	0.147	0.178
Rand	0.004	0.000	0.187	0.159	*0.151*	0.187

signal has therefore *not* degraded, it still ranks nearly all surviving positives into its top 50. What has changed is that the joint positive set (EPSS $> 0.10 \wedge$ HRS $>$ p75) has been patched away by HygienePrio's own selection. Note also that the HRS-only driver drains the HRS upper tail every window, yet HygienePrio holds 0.713 under it, direct evidence against an HRS-tail signal-degradation reading. This reinforces, rather than weakens, the paper's thesis: the collapse is an evaluation artefact of positive-set exhaustion, not an intrinsic scoring failure.

B. EPSS-Self-Driven is the Deepest Sink

The K=200, EPSS-driver column of Table II reads near-zero for every scorer: HP 0.008, EPSS 0.012, HRS 0.002, CVSS 0.001, Random 0.000. The mechanism is the *same* one that produces the HP-driver collapse, exhaustion of the EPSS-gated joint positive set, only deeper: EPSS-only's selection preferentially remediates the highest-EPSS pairs regardless of host state, and by W6 of K=200 the EPSS > 0.10 threshold (Condition 1 of the ground truth) is satisfied by an essentially empty subset of the residual backlog (mean 0.6 surviving positives per backlog, versus 3.8 under T_{HP}). **The positive set itself is exhausted**; no scorer can recover P@50 against an empty positive set.

C. H2 Rejected in Reverse Direction

The pre-registered H2 predicted that EPSS-only's W6 P@50 on its own trajectory would be lower than on the HP-driven trajectory at K=50, reasoning that self-driven EPSS would exhaust its own high-EPSS tail faster. The data show the opposite. At K=50: EPSS-only on T_{EPSS} has W6 P@50 = 0.366; EPSS-only on T_{HP} has W6 P@50 = 0.034. EPSS-only's W6 score is *ten times higher* on its own trajectory than on the HP-driven trajectory. The mechanism is symmetric to C1: when HygienePrio drives, HygienePrio's selection biases the residual backlog *against* EPSS-only's signal (it preferentially patches high-EPSS items that are also on high-HRS hosts, leaving lower-EPSS items behind that are exactly what EPSS-only would have prioritised). Self-driven EPSS keeps its own high-EPSS bias intact through the window-1 selection but does not drain it faster than HP-driven. **H2 is rejected in the opposite direction of the pre-registration.**

TABLE III
PER-PAIR DOMINANCE FRACTION Pr[HP > EPSS] OVER 150 (SEED, WINDOW) PAIRS PER (K, DRIVER).

driver	$K = 50$	$K = 200$
HP	1.000	0.800
EPSS	1.000	0.393
HRS	1.000	1.000
CVSS	1.000	1.000
Rand	1.000	1.000

D. HP Per-Pair Dominance Over EPSS is Trajectory-Conditional

Table III reports the per-pair dominance fraction $\Pr[\mathrm{P@50}^{\mathrm{HP}} > \mathrm{P@50}^{\mathrm{EPSS}}]$ over the 150 (seed, window) pairs per (cell, driver). At K=50, the fraction is 1.000 under every driver. At K=200 the picture is trajectory-conditional: 1.000 under HRS-only, CVSS-only, and Random; 0.800 under HygienePrio-full at this single cell, within the capacity-decay study's observed per-cell range (76.7%-100%, pooled 96.0%); 0.393 under EPSS-only, where both HP and EPSS are near zero. **H3 is rejected at K=200 under T_{EPSS}.**

E. Trajectory Effects Per Scorer

The W6 trajectory-effect range Δ_m^{traj} (max P@50 across drivers minus min) quantifies how trajectory-sensitive each scorer is; Table IV reports it per scorer.

At K=200, HP-full and EPSS-only have the largest trajectory-effect ranges in Table IV, the two scorers whose rankings are most aligned with the EPSS-gated ground-truth positive set, and therefore the two whose measured P@50 collapses hardest when a driver drains that set. CVSS-only and Random have the smallest ranges, their (weak) measured performance depends little on which driver shapes the residual backlog.

TABLE IV
W6 TRAJECTORY-EFFECT RANGE Δ_m^{traj} PER SCORER (MAX MINUS MIN MEAN P@50 ACROSS THE FIVE DRIVERS).

scorer	$\Delta_m^{\text{traj}}(K = 50)$	$\Delta_m^{\text{traj}}(K = 200)$
HP-full	0.242	0.705
EPSS-only	0.375	0.610
HRS-only	0.286	0.318
CVSS-only	0.068	0.178
Random	0.086	0.187

TABLE V
PRE-REGISTERED HYPOTHESIS OUTCOMES.

ID	Decision rule	Outcome
H1	HP W6 collapse at $K = 200$ within ± 0.05 on ≥ 1 non-HP driver	Rejected
H2	EPSS W6 on T_{EPSS} < on T_{HP} at $K = 50$	Rejected
H3	$\Pr[\text{HP} > \text{EPSS}] \geq \{0.75, 0.50\}$ under T_{EPSS}	Rejected

F. Hypothesis Outcomes

Table V summarises the three pre-registered hypothesis outcomes.

G. Pre-Registration Deviations

None. Decision rules and stop rules were applied verbatim. The abstract was rewritten after H1 rejection in compliance with the protocol's stop rule before the discussion was drafted.

VIII. A Theoretical Lower Bound: The Closed-Loop Signal Exhaustion Theorem

The empirical findings of §VII demonstrate that HygienePrio's Precision@50 at $K = 200, w = 6$ depends on which method drives the fleet trajectory: 0.075 under HygienePrio's own driver, 0.701-0.713 under the non-draining drivers (CVSS-only, HRS-only, Random). This section formalises the underlying phenomenon as a theorem applicable to any scoring-driven closed-loop prioritization system.

A. Setup

Let $V_w \subseteq H \times C$ be the unpatched (host, CVE) backlog at the start of window w. Let $S : H \times C \to \mathbb{R}$ be a deterministic scoring policy, and let

$$A_w(S; K) = \{(h, c) \in V_w : S(h, c) \text{ is in the top-}K \text{ of } V_w\} \tag{3}$$

be the policy's top-K action set. The fleet evolves by remediation: $V_{w+1} = V_w \setminus A_w(S; K)$ plus an i.i.d. disclosure draw of size n_w from a fixed distribution $\mathcal{D}_{\text{new}}$.

Let $L \subseteq H \times C$ be the ground-truth positive set at V_w defined by a binary predicate $L_w = \{(h, c) \in V_w : \phi(h, c; F_w) = 1\}$ for some fleet-state-dependent labelling function ϕ. Define the *Precision@K* of S as $P_w(S; K) = |A_w(S; K) \cap L_w|/K$.

B. The Signal-Tail Density Function

Let $\tau(\cdot)$ denote the labelling threshold induced by ϕ under F_w. Define the *signal-tail density* of S at window w as

$$\rho_w(S) = \frac{|\{(h, c) \in V_w : S(h, c) > q_K(S; V_w)\}|}{|V_w|}, \tag{4}$$

where $q_K(S; V_w)$ is the K-th largest value of S on V_w. The density $\rho_w(S)$ measures the fraction of the backlog that lies in S's scoring upper tail.

C. The Theorem

Theorem 1 (Closed-Loop Signal Exhaustion). *Suppose S's top-K action set $A_w(S; K)$ has expected positive-set overlap $\mathbb{E}|A_w(S; K) \cap L_w| = \alpha K$ for some $\alpha \in (0, 1]$ at $w = 1$. Suppose further that:*

- (A1) *The labelling predicate ϕ is positively correlated with S on the upper tail: $\mathbb{P}[\phi(h, c) = 1 \mid (h, c) \in A_w(S; K)] > \mathbb{P}[\phi(h, c) = 1 \mid (h, c) \in V_w \setminus A_w(S; K)]$.*
- (A2) *The new-disclosure rate n_w satisfies $n_w < K$ for all w.*
- (A3) *The new-disclosure distribution $\mathcal{D}_{new}$ has tail density $\rho^{\mathcal{D}}(S) \leq \rho_1(S)$, the disclosure process does not preferentially refill S's upper tail beyond its initial density.*
- (A4) *The initial tail density does not exceed the positive prevalence bound: $\rho_1(S) \leq \alpha$.*

Then for any $\epsilon > 0$, there exists a finite window $w^\star(\epsilon)$ such that for all $w \geq w^\star$,

$$\mathbb{E}[P_w(S; K)] \leq \mathbb{E}[P_w(\text{Random}; K)] + \epsilon. \tag{5}$$

That is, the closed-loop policy converges in expectation to the random-ranking baseline within finite windows.

D. Proof Sketch

Let $T_w(S) = \{(h, c) \in V_w : S(h, c) > q_K(S; V_w)\}$ denote the upper-tail set on V_w. Note that T_w is redefined on each window's backlog V_w; the induction below is over the surviving positive mass within the (re-anchored) tail, not over a fixed set. By A1, T_w is enriched in positives: $\mathbb{P}[\phi = 1 \mid T_w] > \mathbb{P}[\phi = 1 \mid V_w]$.

Each window's selection $A_w \subseteq T_w$ removes positives at a rate at least proportional to the enrichment factor: $\mathbb{E}|A_w \cap L_w| \geq \alpha K$. The replacement disclosure n_w contributes new positives at rate $\mathbb{P}_{\mathcal{D}_{\text{new}}}[\phi = 1] \cdot n_w$, which by A3 is bounded above by $\rho^{\mathcal{D}}(S) \cdot n_w \leq \rho_1(S) \cdot n_w \leq \alpha \cdot n_w < \alpha K$, using A3, A4, and A2 in turn.

The net change in positive mass within T_w per window is therefore negative:

$$\Delta|T_w \cap L_w| \leq -\alpha K + \rho^{\mathcal{D}}(S) \cdot n_w < 0. \tag{6}$$

By induction, $|T_w \cap L_w| \to 0$ as $w \to \infty$. Since $A_w \subseteq T_w$, $P_w(S; K) = |A_w \cap L_w|/K \to 0$. As the random baseline $P_w(\text{Random}; K) \to \mathbb{P}[\phi = 1 \mid V_w]$ also approaches zero (or its limiting positive density), $|P_w(S; K) - P_w(\text{Random}; K)|$ becomes arbitrarily small.

As an order-of-magnitude heuristic (not a tight bound), the exhaustion window scales as $w^\star \approx \lceil |T_1 \cap L_1|/(\alpha K - \rho^{\mathcal{D}}(S) n_w) \rceil$, the initial positive tail mass divided by the net per-window drain; the exact constant depends on the gap between αK and $\rho^{\mathcal{D}}(S) n_w$. □

E. Empirical Consistency

The data of §VII show ceiling-limited precision driven by positive-set exhaustion, the mechanism the theorem formalizes. At $K = 200, \lambda = 3$ under the self-driven trajectory, the mean surviving positive mass at $w = 6$ is 3.8 of $\approx$ 760 backlog pairs, capping P@50 at 0.076; HygienePrio's measured 0.075 sits at that ceiling. Measured precision nonetheless remains well above the random baseline (0.075 vs. 0.004), because the evaluation truncates at W6 before full convergence to the theorem's conclusion $\mathbb{E}[P_w(S;K)] \leq \mathbb{E}[P_w(\text{Random};K)] + \epsilon$. Under the non-draining drivers the labelling tail is not preferentially drained, the exhaustion dynamic is absent, and $P_w(S;K)$ remains near 0.7.

The theorem is likewise consistent with the EPSS-self-driven collapse to 0.001-0.012: EPSS-only's selection at $K = 200$ drains the same EPSS-gated positive set, and the disclosure rate $\lambda = 3$ contributes few high-EPSS pairs per window, so the net drain rate is sharp and $w^\star$ is small.

F. Implication

Theorem 1 formalises a structural limit: *no deterministic scoring policy whose selection is positively correlated with its own discriminative signal can sustain Precision@K indefinitely under capacity-constrained closed-loop deployment.* The mechanism is not particular to HygienePrio; it applies equally to EPSS-only, CVSS-only, or any future scoring rule.

The practical interpretation matches the operational recommendation of §IX: *decouple scoring from selection at high capacity.* The theorem provides the underlying reason: a self-driven policy is mathematically guaranteed to exhaust its own signal in finite time; a foreign-driven policy is bounded only by the foreign driver's exhaustion rate of the labelling tail, which can be slower or non-existent.

G. Corollaries

The theorem has four corollaries that strengthen its operational reach.

Corollary 1 (Random Policy Escape)**.** *If S is the random ranking baseline, then $A_w(S;K)$ is a uniform random sample of V_w, the enrichment assumption A1 is violated by construction, and $P_w(S;K) = \mathbb{P}[\phi = 1 \mid V_w]$ for all w. Random ranking does not exhibit signal exhaustion.*

Corollary 2 (ϵ-Greedy Policies Slow Convergence)**.** *For $\epsilon \in (0,1)$, consider the mixed policy S_ϵ that selects each window's top-K from S with probability $1-\epsilon$ and uniformly at random from V_w with probability ϵ. Then the convergence rate of Theorem 1 is slowed by a factor of $1/(1-\epsilon)$, approximately, by the following heuristic argument. The expected positive-mass drain per window becomes $(1-\epsilon)\alpha K + \epsilon\mathbb{P}[\phi = 1 \mid V_w]K$, which approaches the random baseline as $\epsilon \longrightarrow 1$. The signal-exhaustion regime is asymptotically equivalent but reached in $w^\star_\epsilon = w^\star/(1-\epsilon)$ windows.* □

Corollary 3 (High Arrival Rates Break the Theorem)**.** *If A2 is violated ($n_w \geq K$) and A3 is violated in the same direction ($\rho^{\mathcal{D}}(S) \geq \alpha$), the net change in positive mass within T_w becomes non-negative and the theorem does not apply. Empirically, this regime corresponds to a fleet whose new-CVE arrival rate exceeds its remediation rate while preserving the upper-tail composition, i.e., a disclosure process biased toward exactly the signals the policy uses to rank. In the simulated $\lambda = 12, K = 10$ cell of the capacity-decay study, this condition is approached but not crossed; in real fleets it would correspond to a denial-of-service attack on the prioritization system (deliberate publication of many high-EPSS CVEs targeting the defender's fleet topology).*

Corollary 4 (Online Calibration Cannot Escape the Bound)**.** *If S is replaced at each window by a re-calibrated S_w trained on a held-out calibration set whose backlog V_w^{cal} also evolves under the same closed-loop policy, then S_w's discriminative target is the same evolving distribution as S's; the calibration procedure cannot move the labelling threshold τ faster than the positive-mass drain rate. Online recalibration therefore inherits Theorem 1's convergence bound asymptotically.* This explains the online-calibration study's empirical finding that lag-1 online recalibration cannot rescue K=200 collapse*: the calibration target is itself exhausted by the closed-loop policy. The only structural way to escape the bound is to make the labelling function depend on signals* outside *the policy's discriminative set, which is exactly what self-trajectory decoupling accomplishes.*

H. Scope and Limitations

The theorem assumes deterministic scoring, bounded new-disclosure rate, and a labelling predicate positively correlated with S on the upper tail. The four corollaries above cover the randomised, ϵ-mixed, high-arrival, and online-recalibrated extensions. The theorem does not cover:

- Disclosure processes that adversarially refill S's upper tail faster than K (Assumption A3 fails). The 2026 DBIR's ~43-day mean patch-to-disclosure cycle [4] suggests this is unlikely in practice outside the deliberate adversarial-disclosure scenario of Corollary 3.
- Reinforcement-learning policies whose scoring updates between windows, which require a different bound for the update-cycle dynamics.

Extending the theorem to these cases is left as future work.

IX. Discussion

A. What This Paper Changes About the capacity-decay study

The capacity-decay study [1] reported HygienePrio's H4 absolute-floor collapse at $K \in \{100, 200\}$ as a property of hygiene-augmented scoring under high capacity. The present paper shows the collapse is substantially a property of *positive-set-draining deployment*: when the queue-driving policy preferentially remediates the EPSS∧HRS joint positive set, that set exhausts, and every scorer aligned with it is capped by the surviving positive mass (§VII: 3.8 mean positives, ceiling 0.076). Under the non-draining drivers, CVSS-only, HRS-only, and the trivial Random baseline, HygienePrio's W6 mean P@50 at K=200 lands at 0.701-0.713, an order of magnitude above the self-trajectory number. EPSS-driven selection, by contrast, also collapses HygienePrio's measured P@50 (0.008) because it drains the same EPSS-gated positive set.

The operational corollary is sharp. A team that uses HygienePrio for scoring and lets some *other* policy, for example, a ticketing system, a compliance-driven schedule, or a CVSS-based queue, determine which pairs are actually patched first will not experience the capacity-decay study collapse. The collapse is a closed-loop artefact, not a property of hygiene-augmented scoring as a method.

B. Why EPSS-Self-Driven Destroys Everything

The K=200, T_{EPSS} column of Table II contains the most extreme cell in the grid. Five W6 P@50 values, all in $[0.000, 0.012]$. The mechanism is ground-truth-set exhaustion: EPSS-only's selection at K=200 removes the high-EPSS tail so fast that the binary positive condition (EPSS > 0.10) is satisfied by an essentially empty residual backlog. Once the positive set is empty, P@50 is mechanically zero regardless of which scorer is being evaluated, you cannot rank a non-existent positive.

This is the same mechanism that produces the HP-driven collapse, taken to its extreme. Both HygienePrio-driven and EPSS-driven selection drain the EPSS-gated joint positive set; EPSS-only drains it deeper (mean 0.6 surviving positives versus 3.8 under T_{HP}), reaching a literal zero floor. Both are positive-set-exhaustion artefacts of the evaluation; neither is an intrinsic scoring failure.

C. What the capacity-decay study's 96% Per-Pair Dominance Actually Means

The capacity-decay study reported that HygienePrio-full > EPSS-only at P@50 in 96.0% of (cell, seed, window) triples (2,881/3,000, pooled over its 20-cell grid) under the (then-implicit) HygienePrio-driven trajectory (the temporal-stability study's own single-cell figure was 150/150). The present paper finds that this statistic survives at $K = 50$ under every driver (always 1.000), survives at $K = 200$ under HRS, CVSS, and Random drivers (1.000), partially survives at $K = 200$ under HygienePrio's own driver (0.800), and *does not* survive at $K = 200$ under the EPSS-only driver (0.393). The interpretation: per-pair dominance is a robust property of HygienePrio under any non-draining selection, including the natural baselines an operations team would deploy if not using HygienePrio's score for selection. Under the EPSS-self-driven regime both methods collapse together, and the comparison becomes degenerate.

D. What the capacity-decay study's H4 Rejection Means After This Paper

The capacity-decay study's H4 rejection now reads as follows. HygienePrio-full's absolute W6 P@50 at $K \geq 100$ is degraded only when the trajectory is driven by a policy that drains the EPSS$\wedge$HRS joint positive set (HygienePrio itself, or EPSS-only). The mechanism is joint-positive-set exhaustion. On any non-draining driver, the absolute floor is intact at the temporal-stability study level (~ 0.5 or higher). The original H4 hypothesis "HygienePrio retains ≥ 0.4 at every cell" is *re-supportable* under the qualified statement "on every cell, on at least one non-draining driver". We do not formally re-test the capacity-decay study H4 here because doing so would constitute hypothesising after results known (HARKing); we report the observation honestly and leave a new pre-registered H4-style restatement to future work.

E. Why the Trajectory Effect is Asymmetric Across Scorers

The trajectory-effect range Δ_m^{traj} is largest for HP-full and EPSS-only and smallest for CVSS-only and Random. The asymmetry has a clean structural interpretation: a scorer's W6 performance depends on whether the driver preferentially drained the ground-truth positive set that the scorer's ranking is aligned with. HP-full and EPSS-only rank the EPSS-gated joint positive set highest, so the HP-driven and EPSS-driven trajectories, both of which drain that set, collapse their measured precision. CVSS-only relies on the CVSS upper tail (which is not part of the ground-truth condition) so no driver in the grid preferentially starves it. Random relies on no signal. The two scorers most sensitive to trajectory are the two scorers whose ranking is most aligned with the ground-truth positive set, which is the same property that makes them the strongest scorers in the first place.

F. Operational Recommendations

Sharpened from Papers 5-8:

- Use HygienePrio for scoring across all studied capacity regimes; its per-pair dominance over EPSS-only is preserved whenever the patching queue is driven by a policy that does not drain the EPSS$\wedge$HRS joint positive set. CVSS-, HRS-, or compliance-driven queues qualify; HygienePrio-driven and EPSS-driven queues at high capacity do not.
- Do not let HygienePrio be both the scorer and the queue-driving policy at $K \geq 100$. Either (i) decouple, use HygienePrio for ranking but let another policy (ticketing system, CVSS queue, human triage) drive the patching cycle, or (ii) accept the capacity-decay study H4-style collapse as the cost of full automation.
- Do not deploy EPSS-only as both scorer and selection policy at high capacity. EPSS-self-driven at K=200 produces a literal zero floor by W6.

G. Relationship to the Research Sequence

This paper closes a gap that Papers 5-8 explicitly bounded but did not measure. The cumulative arc is unchanged in spirit: hygiene augmentation helps prioritization. But the headline "HygienePrio holds up across regimes" is now sharper. It holds up across regimes *except* when fleet evolution at high capacity is driven by a positive-set-draining policy, HygienePrio itself, or EPSS-only. That last condition is operationally controllable, and the recommendation is to control it.

X. Threats to Validity

The threats taxonomy follows Wohlin et al. [5].

Conclusion validity. The 7,500-row evaluation supports $\sim$ 50-row cell-means per (K, driver, scorer, window), a reasonably tight estimator. Bootstrap CIs (10,000 resamples) at the W6 cells separate the HP-driven HP value (0.075) from the CVSS/HRS/Random-driven HP values (0.701-0.713) without overlap. The qualitative finding does not depend on the CI width.

Internal validity.

(i) Per-window ground truth re-uses HRS percentiles. The binary positive condition incorporates HRS > 75th percentile recomputed each window. Under HP-driven and HRS-driven trajectories, the recomputed percentile threshold moves substantially each window; under CVSS-driven and Random-driven trajectories, it moves less. This means the "HP scored on T_CVSS" cell is operating against a relatively stable positive set whose composition is similar to W1, while the "HP scored on T_HP" cell is operating against a fast-moving positive set. The non-identifiability is unavoidable under the per-window threshold convention; we report it transparently and bound claims to that convention.

(ii) The EPSS-self-driven collapse is partly a structural artefact. When EPSS-only's selection drives evolution at K=200, nearly all CVEs with EPSS > 0.10 are remediated by W3-W4, and the ground-truth positive condition $EPSS > 0.10$ becomes near-empty. The literal zero W6 P@50 values reflect an essentially empty positive set, not a scoring failure. We report this honestly; the qualitative finding ("EPSS-self-driven is the deepest sink") is robust but should not be read as "EPSS-only is the worst scorer".

(iii) Two cells only. K=50 and K=200 anchor low and collapse regimes; intermediate capacities (e.g., K=100) are not measured here. The capacity-decay study H4 dataset can be used by readers to fill in the K=100 cell under HP-driven trajectory.

Construct validity. P@50 with binary relevance, recomputed-per-window HRS-percentile ground truth, and the HygienePrio scorer's fixed weights are inherited from Papers 5-8. We deliberately do not vary calibration strategy (the Papers 7-8 axis) because doing so would interact with the trajectory question; isolating one effect at a time is a methodological choice that the protocol locks.

External validity. Inherited from Papers 4-8: synthetic EEHDA structural priors, 14-day window, 0.15 patch debt reduction, 0.02 EPSS random walk, 8% KEV prevalence. The qualitative finding, that recursive-deployment collapse is selection-induced, is more robust than any specific number; the specific W6 levels (0.075 vs 0.701) are distributional. External validation on real fleet telemetry with longitudinal exploitation outcomes is the appropriate next step.

XI. Related Work

Trajectory-coupling threat in Papers 5-8. Our temporal-stability study (§9), the capacity-decay study [1] §9, the online-calibration study (§9), and the multi-history smoothing study (§9) all acknowledge the HygienePrio-driven trajectory as an internal-validity threat. Each cited a "fully factorial each-method-drives-its-own" future-work item as the appropriate counterfactual. The present paper implements that counterfactual in a pre-registered form.

High-capacity collapse in the capacity-decay study. The capacity-decay study reported HygienePrio-full's W6 P@50 collapse at $K \in \{100, 200\}$ and attributed it to high-HRS-tail exhaustion. The present paper retains the collapse statistic on the HP-driven trajectory but shows it disappears under the non-draining drivers (CVSS-only, HRS-only, Random), revising the interpretation from "intrinsic property of hygiene-augmented scoring at high capacity" to "closed-loop artefact of recursive deployment".

Vulnerability scoring and prioritization. Probabilistic and ordinal scorers for vulnerability triage span the deterministic severity of CVSS [12], the stakeholder-specific decision trees of SSVC [13], and exploitation-likelihood scoring grounded in case-control studies of which vulnerabilities are actually attacked [14]. HygienePrio sits in this lineage as a probabilistic scorer; the self-trajectory question we raise, whether a scorer's measured quality is intrinsic or trajectory-induced, applies to any of these scorers once it is also used to drive remediation.

EPSS dynamics. Jacobs et al. [6], [7] established per-CVE exploit-prediction scoring. Ravalico et al. [8] studied per-CVE EPSS score drift. The present paper documents a distinct effect: *EPSS-as-policy* remediation at high capacity exhausts the ground-truth positive set within six windows on the simulated fleet, producing a literal zero floor for every scorer evaluated against that trajectory. Neither prior study addressed the policy-level effect.

Counterfactual evaluation in security ML. The general pattern, evaluating method A on the data trajectory that method B would have generated, appears in counterfactual learning to rank and off-policy evaluation. The instance studied here is the simplest form: a deterministic policy generates the trajectory, and the counterfactual is computed by re-running the simulation under each candidate policy. We do not survey the broader counterfactual-policy literature; the question this paper resolves is operationally specific to the capacity-decay study H4 claim.

Cyber-hygiene benchmarking. The HygieneBench synthetic benchmark, HygienePrio [2], the temporal extension, the capacity sweep [1], online calibration, and multi-history calibration together constitute the methodological substrate within which this paper sits.

Policy mandates. CISA BOD 22-01 [9] and 23-01 [10] along with NIST SP 800-40 Rev. 4 [11] define the operational scheduling regime under which any vulnerability-prioritization deployment must operate. The recommendation that scoring and selection should be decoupled at high capacity follows from this paper's findings and aligns with the spirit of the BOD mandates' separation of discovery, prioritisation, and remediation roles.

XII. Conclusion

A pre-registered 7,500-row self-trajectory evaluation of the five HygienePrio-family prioritization methods at $K \in$

$\{50, 200\}$, $\lambda = 3$, $W = 6$, on the 25 evaluation seeds inherited from Papers 5-8, produces three findings that retroactively reframe the research sequence.

First, the capacity-decay study's headline "HygienePrio collapses at high capacity" is rejected as an intrinsic property of hygiene-augmented scoring. At $K = 200, w = 6$, HygienePrio-full's mean P@50 is 0.075 on its own trajectory but 0.701 on CVSS-driven, 0.713 on HRS-driven, and 0.706 on Random-driven trajectories. The collapse is a positive-set-exhaustion artefact: HygienePrio's own selection patches away the EPSS$\wedge$HRS joint positive set, leaving a mean of 3.8 positives per backlog, a P@50 ceiling of 0.076 that HygienePrio's still-intact signal nearly attains (0.075, versus Random's 0.004).

Second, EPSS-only's selection at $K = 200$ produces a literal zero W6 P@50 for every scorer (range 0.001-0.012). The ground-truth positive set is structurally exhausted by EPSS-only's aggressive remediation of the high-EPSS tail. EPSS-self-driven at high capacity is the deepest sink in the studied grid.

Third, HygienePrio's per-pair dominance over EPSS-only is trajectory-conditional. At K=50 it is 1.000 under every driver; at K=200 it is 1.000 under HRS, CVSS, and Random drivers, 0.800 under HP's own driver, and 0.393 under EPSS-only's driver (where both methods collapse together). The capacity-decay study's pooled 96% headline statistic (the temporal-stability study's own single-cell figure was 150/150) is preserved at moderate capacity and under non-draining drivers, the regimes most relevant to actual deployment.

The operational reading is sharper than Papers 5-8 allowed: HygienePrio is a strict improvement over EPSS-only across the studied regimes whenever the patching queue is driven by a policy that does not drain the EPSS$\wedge$HRS joint positive set. CVSS-, HRS-, or compliance-driven queues qualify; HygienePrio-driven and EPSS-driven queues at high capacity do not (HygienePrio's dominance falls to 0.393, and its measured floor to 0.008, under T_{EPSS} at $K = 200$). The capacity-decay study H4 collapse is a known closed-loop hazard, not a property of the scoring method itself, and is operationally controllable by driving the queue with a non-draining policy.

Reproducibility. The runner, analysis script, preregistration protocol, and frozen 7,500-row results CSV are available at https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper9.

Future work. *(i)* A pre-registered re-test of the capacity-decay study's H4 hypothesis under the qualified statement "on every cell, on at least one non-draining driver". Doing this rigorously requires a fresh seed set to avoid HARKing on the present data. *(ii)* Trajectory effects on calibration: do the online-calibration study's lag-1 recovery ratio and the multi-history smoothing study's smoothing penalty change under non-HP drivers? Both papers measured under T_HP only. *(iii)* External validation on real fleet telemetry with longitudinal exploitation outcome data. *(iv)* A cross-trajectory evaluation in which the scorer *also* predicts the next-window driver, modelling a fully adaptive closed-loop policy rather than two static ones.

Data Availability Statement

The data and code that support the findings of this study are openly available in the research-papers repository at https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper9. The manuscript sources, figures, analysis code, and frozen result tables are included and regenerate every reported number deterministically from the fixed seeds.

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